BITS Pilani • Work Integrated Learning Programmes (WILP)

M.Tech (AIML) — Machine Learning (Assignment - 2)

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Programme: M.Tech in AI & ML	**Topic:** Multi-Model Classification & Streamlit Cloud

1. Mandatory Submission Links

Deliverable	Clickable URL / Submission Link
GitHub Repository Link *(Source code, requirements.txt, README.md, test_data.csv)*	https://github.com/your-username/bits-ml-assignment-2
Live Streamlit App Link *(Deployed on Streamlit Community Cloud)*	https://bits-ml-assignment-2.streamlit.app

2. Proof of Execution on BITS Virtual Lab [1 Mark]

As per assignment instructions, execution on the BITS Virtual Lab environment is verified below:

[BITS VIRTUAL LAB SCREENSHOT PLACEHOLDER]

Take a screenshot of your assignment execution inside BITS Virtual Lab and save it as:
bits_lab_screenshot.png *in this project folder, then re-run:*
python generate_submission_pdf.py
It will automatically embed your screenshot here.

3. GitHub README Content (Assignment Details & Results)

a. Problem Statement

In modern digital commerce, understanding customer browsing behavior and accurately predicting whether an online session will culminate in a purchase transaction is critical for revenue growth, dynamic personalized marketing, and conversion rate optimization (CRO). The goal of this project is to develop, evaluate, and deploy an end-to-end Machine Learning classification system that predicts Online Shoppers Purchasing Intention (Revenue: True/False). Utilizing a multi-model comparative framework, we implement, benchmark, and deploy five distinct classification algorithms: Logistic Regression, Decision Tree Classifier, K-Nearest Neighbor (KNN) Classifier, Naive Bayes Classifier (Gaussian NB), and Support Vector Machine (SVM with RBF Kernel). The models are evaluated across six key metrics: Accuracy, AUC-ROC, Precision, Recall, F1 Score, and Matthews Correlation Coefficient (MCC).

b. Dataset Description [1 Mark]

Repository: UCI Machine Learning Repository / Kaggle
Dataset: Online Shoppers Purchasing Intention Dataset
Total Instances: 12,330 sessions (satisfies requirement: minimum 500)
Total Features: 17 input attributes + 1 target attribute = 18 total (satisfies requirement: minimum 12 features)
Classification Type: Binary Classification
Target Variable: Revenue (True: 15.47%, False: 84.53%)
Key Attributes: Administrative, Administrative_Duration, Informational, Informational_Duration, ProductRelated, ProductRelated_Duration, BounceRates, ExitRates, PageValues, SpecialDay, Month, OperatingSystems, Browser, Region, TrafficType, VisitorType, Weekend.

c. Models Used & Evaluation Comparison Table [5 Marks]

All 5 models were trained on the identical 80% training split (9,864 samples) and tested on the 20% holdout split (2,466 samples) with standardized ColumnTransformer preprocessing:

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8808	0.8828	0.7366	0.3586	0.4824	0.4592
Decision Tree *[Winner]*	**0.8994**	**0.9178**	0.7107	0.5916	**0.6457**	**0.5910**
KNN	0.8751	0.8101	0.6907	0.3508	0.4653	0.4327
Naive Bayes	0.7810	0.8065	0.3799	**0.6545**	0.4808	0.3752
Support Vector Machine	0.8873	0.8824	0.7047	0.4686	0.5629	0.5149

d. Observations on Model Performance [3 Marks]

ML Model Name	Observation about model performance
Logistic Regression	Solid linear baseline with high accuracy (88.08%) and high precision (73.66%), but low recall (35.86%) due to inability to model non-linear session threshold interactions.
Decision Tree	Best overall classifier. Captures hierarchical visitor engagement rules (e.g. PageValues thresholds) achieving top AUC (0.9178), top F1 (0.6457), top MCC (0.5910), and top Accuracy (89.94%).
KNN	Decent precision (69.07%) and 87.51% accuracy, but suffers from metric sparsity in high-dimensional one-hot encoded space resulting in low recall (35.08%) and lower AUC (0.8101).
Naive Bayes	Demonstrates high recall (65.45%), effectively discovering converters, but suffers from low precision (37.99%) and lower accuracy (78.10%) because the feature independence assumption fails on correlated session variables.
Support Vector Machine	Strong runner-up with 88.73% accuracy and 0.8824 AUC. The RBF kernel captures non-linear boundary separation well, maintaining 70.47% precision and 0.5149 MCC.

Overall Winner for the Dataset: Decision Tree Classifier

The Decision Tree classifier is the indisputable winner across AUC (0.9178), F1-Score (0.6457), Matthews Correlation Coefficient (0.5910), and Accuracy (89.94%). Web browsing and purchase intent behavior inherently follow sequential, condition-based decision steps (e.g., *if PageValues > threshold AND BounceRates < limit*), which tree-based architectures naturally represent with high interpretability and minimal distortion.